LACTULOSE

Hepatic encephalopathy

Takes 1–2 days

CHEMICAL NAME 4-O-β-D-galactopyranosyl-D-fructose (synthetic disaccharide)
CLASS Osmotic laxative; ammonia-lowering agent
TRADE NAMES Duphalac · Lactulose solution · Kristalose · Enulose

THERAPEUTIC USES / INDICATIONS

- Constipation (chronic, opioid-induced, in pregnancy, children).
- Hepatic encephalopathy (treatment and prevention).

ADMINISTRATION

- **PO** solution; may mix with water or juice.
- **Rectal** retention enema (hepatic coma).
- Takes 48 h for laxative effect.

DOSAGES

- **Constipation (adult):** start 15–45 mL/day, maintenance 15–30 mL/day (once daily or 2 divided doses).
- **Hepatic encephalopathy:** 30–45 mL three or four times daily, titrated to 2–3 soft stools/day.
- **Children (daily dose):** <1 yr up to 5 mL; 1–6 yr 5–10 mL; 7–14 yr start 15 mL, maintenance 10–15 mL.

PHARMACOKINETICS

- **Onset:** 24–48 h.
- Not absorbed; metabolised by colonic bacteria to lactic and acetic acids.
- Excreted in faeces.

PHARMACODYNAMICS

- Osmotic effect draws water into colon → softer, bulkier stools.
- Acidifies colon → converts NH_3 to non-absorbable NH_4^+ and promotes ammonia-consuming flora → ↓ blood ammonia.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Flatulence, bloating, abdominal cramps.
- Nausea; very sweet taste.
- Diarrhoea (excess dose).

ADVERSE EFFECTS (SERIOUS)

- Dehydration, hypernatraemia, hypokalaemia (excessive diarrhoea, especially in encephalopathy).

CONTRAINDICATIONS & CAUTIONS

- Galactosaemia.
- Intestinal obstruction.
- **Caution:** lactose intolerance, diabetes (contains some sugars).

KEY INTERACTIONS

- **Antacids, neomycin:** may ↓ effect in encephalopathy.
- Diarrhoea ↑ risk with other laxatives.

NURSING MANAGEMENT

- Record stool frequency and consistency.
- Encephalopathy: titrate to 2–3 soft stools; monitor mental status, ammonia, electrolytes.
- Make sure fluid intake.
- Monitor for dehydration.

PATIENT EDUCATION

- Effect takes 1–2 days.
- Bloating usually settles; mix with juice to mask taste.
- Drink plenty of fluids; increase fibre and activity.
- Liver patients: aim for 2–3 soft motions a day — do not stop.

KEY POINTS

- Osmotic laxative; onset 24–48 h.
- Flatulence, cramps.
- Hepatic encephalopathy: titrate to 2–3 soft stools/day.
- Contraindicated in galactosaemia.

SOURCES CHECKED **[551]** Duphalac 3.335 g/5 ml Oral Solution (UK SmPC) **[552]** Lactulose 10 g / 15 ml oral solution sachets (UK SmPC) **[553]** Generlac, oral, rectal (US label) **[554]** Lactulose, oral, rectal (US label)
Full references p. 432–457